On-Board RFI detection

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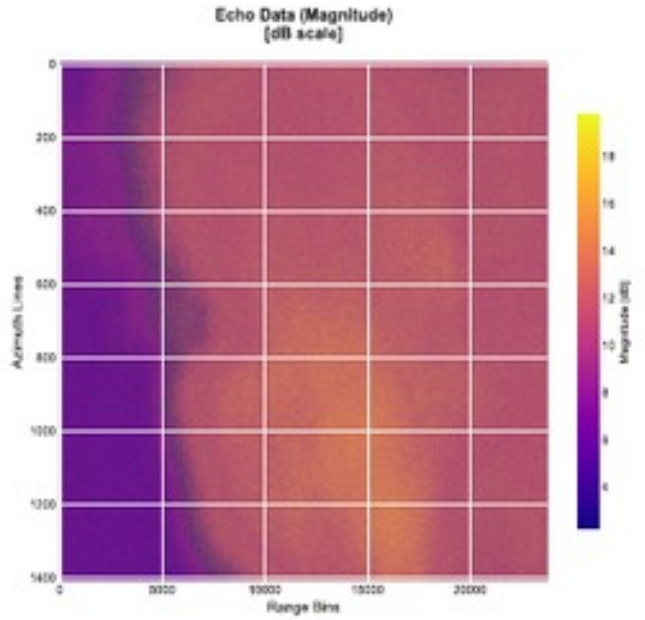
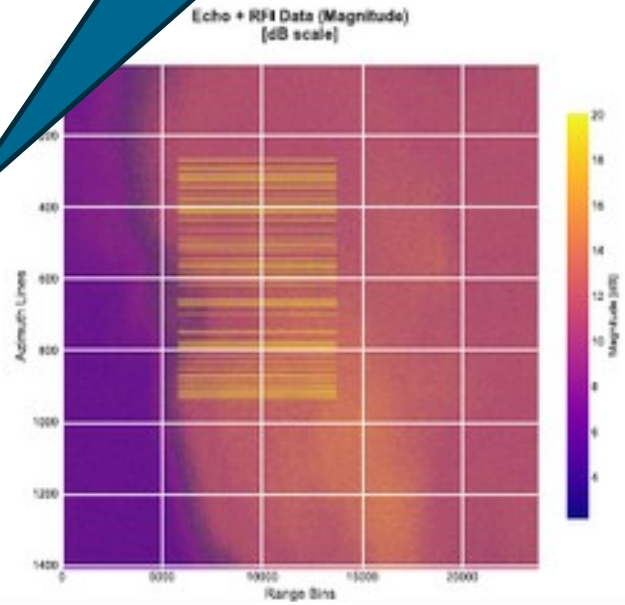
Background

Radio Frequency Interference (RFI)

- Signals human generated in the same frequency used
- Dove avvengono
- Don’t allow to see and ruins the SAR image
- Synthetic Aperture Radar

Riusciamo ad avere due immagini di migliore risoluzione? Magari le prendiamo dal dataset e le plottiamo bene una affianco all’altra in un’unica figura

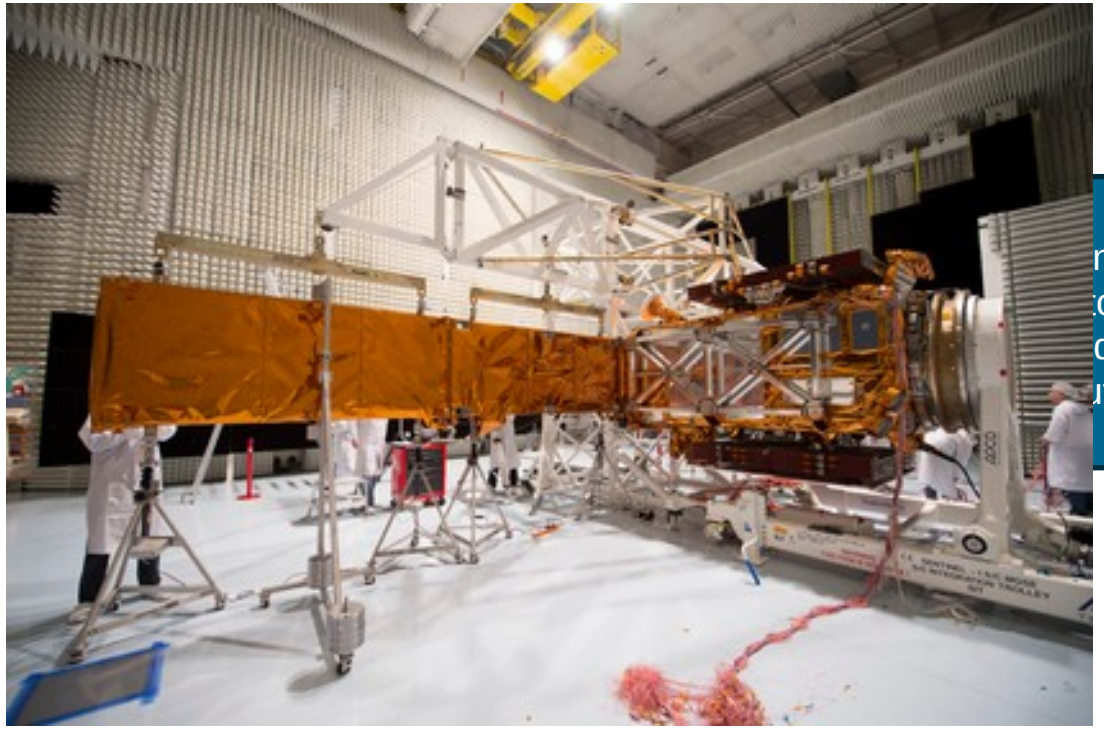




Background

Problems

- Data is hard to find and no ground truth exists in those cases
- Lightweight/Efficency in the model is necessary



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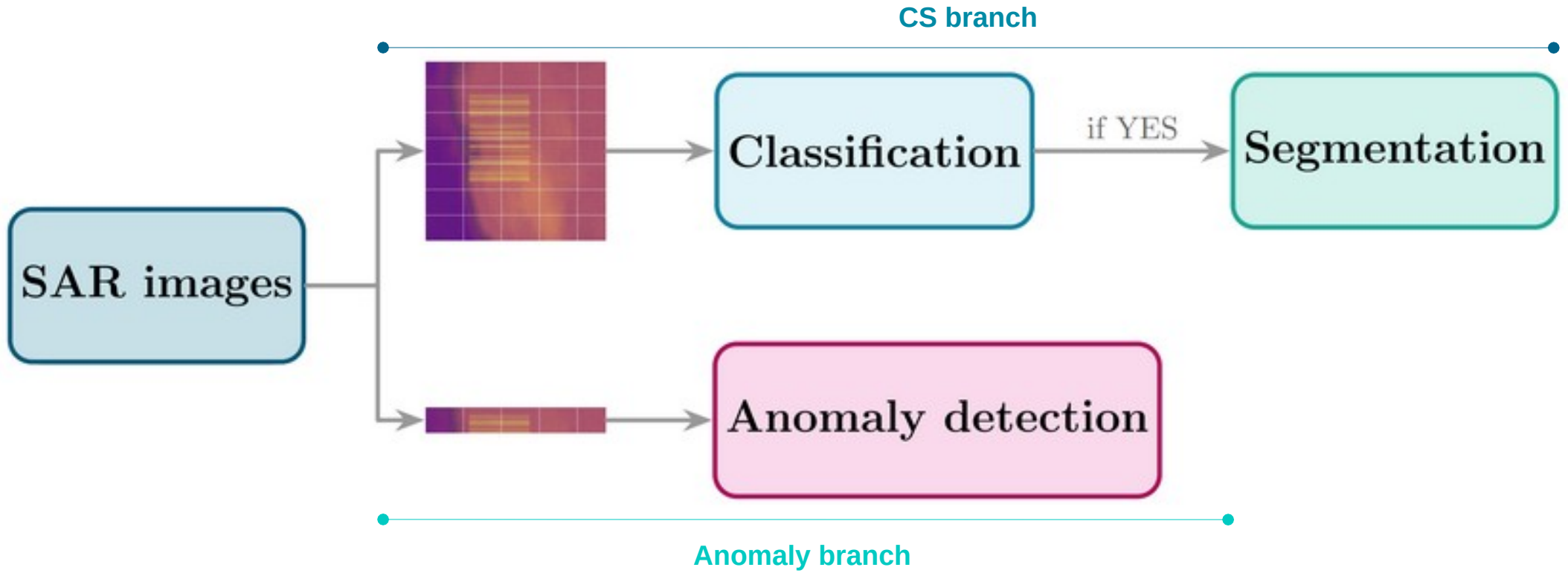
Proposed approach

Two branch pipeline

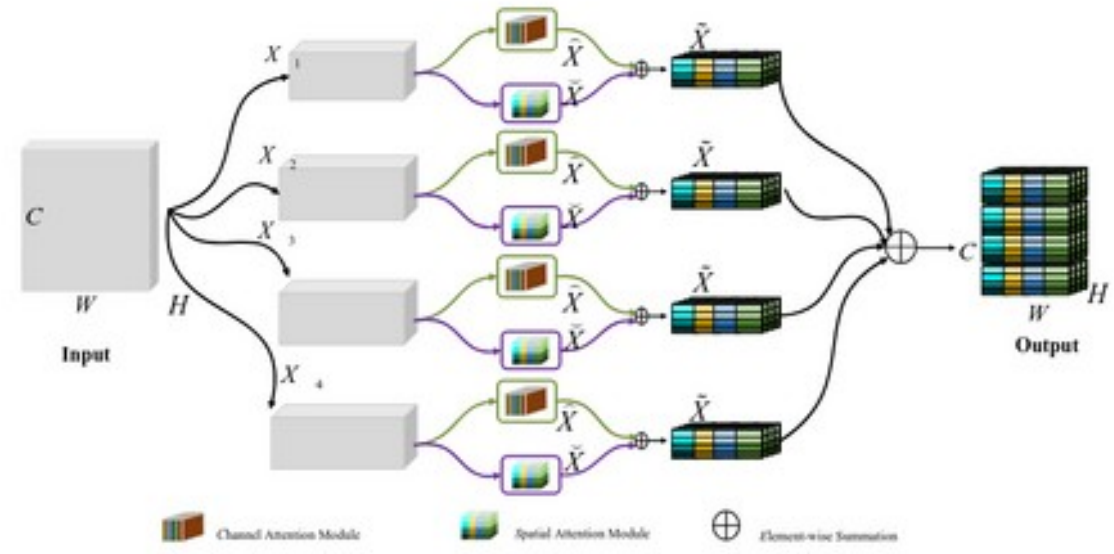


- **CS brach:** first a classification step to see if the RFI is there then segmentation to find it
- **Anomaly branch:** check for anomaly on th burst scale

CS brach - Classification

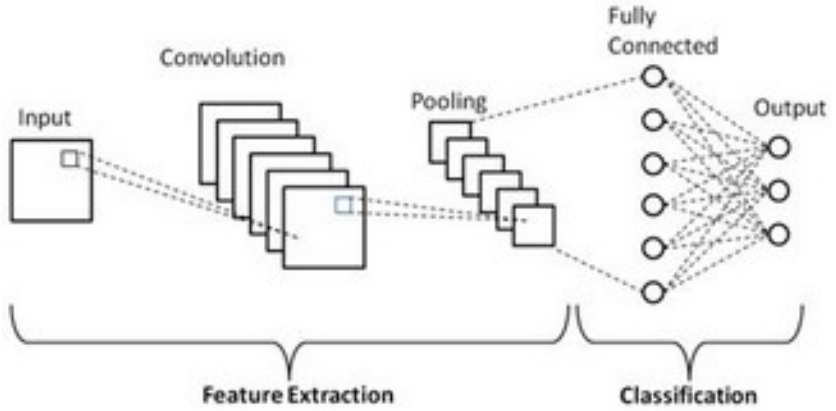
Models

Unconditioned DDPM Trained From Scratch [1]



- lightweight
- Still achives SOTA scores

Simple CNN [2]



- Baseline
- Even lighter and more customizable

[1]Shi, C., Zhang, X., Sun, J., & Wang, L. (2022). A lightweight convolutional neural network based on group-wise hybrid attention for remote sensing scene classification. Remote Sensing, 14(1), Article 161. https://doi.org/10.3390/rs14010161

CS brach - Classification

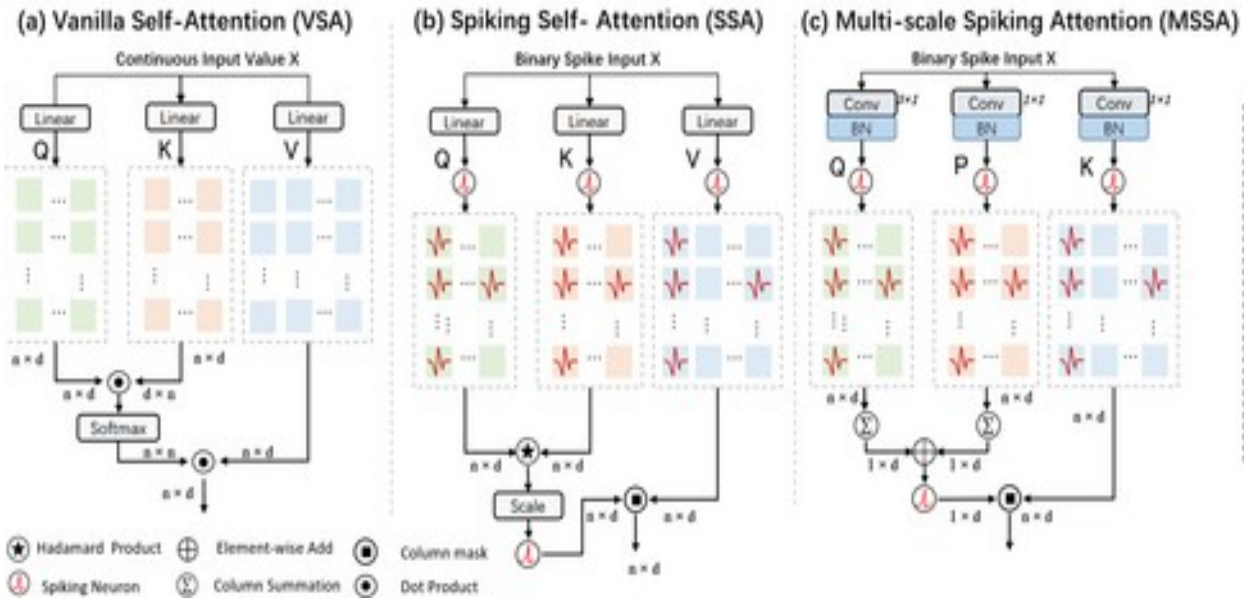
Results

Method	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
DDPM [1]	…			
CNN [2]	…			

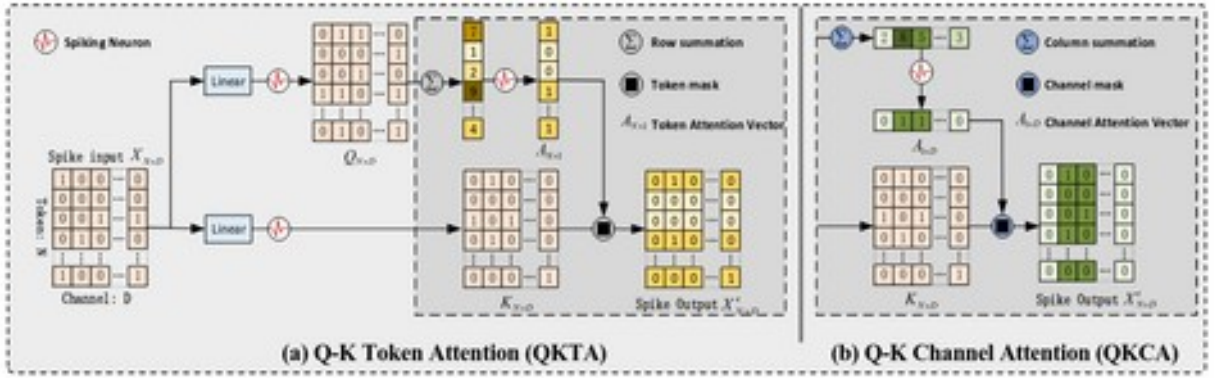
CS brach - Segmentation

Models

MSViT [3]



QK attention [4]



- Bello perché 1
- Bello perché 2

[3]Hua, W., Zhou, C., Wu, J., Chua, Y., & Shu, Y. (2025). MSVIT: Improving spiking vision transformer using multi-scale attention fusion. ArXiv. https://arxiv.org/abs/2505.14719v3

[4]Zhou, C., Zhang, H., Zhou, Z., Yu, L., Huang, L., Fan, X., Yuan, L., Ma, Z., Zhou, H., & Tian, Y. (2024). QKFormer: Hierarchical spiking transformer using Q-K attention. arXiv.

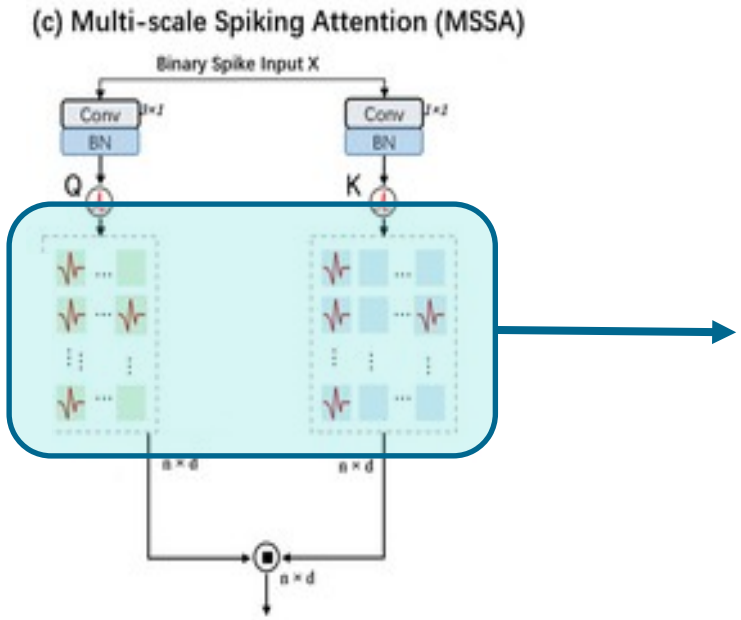
CS brach - Segmentation

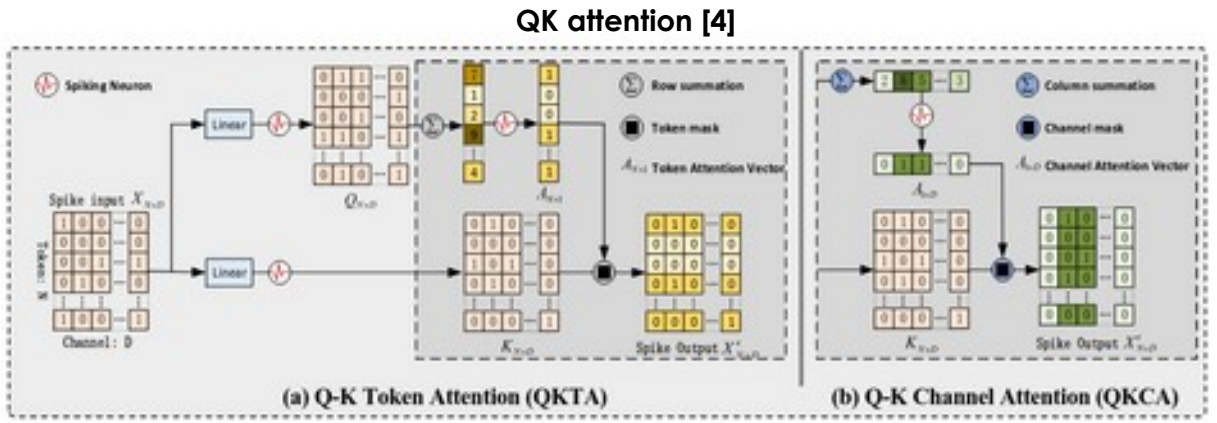
Results

Method	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
DDPM [1]	…			
CNN [2]	…			

CS brach - Segmentation

Proposed approach





- Puts together the power of MSViT with the efficiency of QKformer

[4]Zhou, C., Zhang, H., Zhou, Z., Yu, L., Huang, L., Fan, X., Yuan, L., Ma, Z., Zhou, H., & Tian, Y. (2024). QKFormer: Hierarchical spiking transformer using Q-K attention. arXiv.

CS brach - Segmentation

Results

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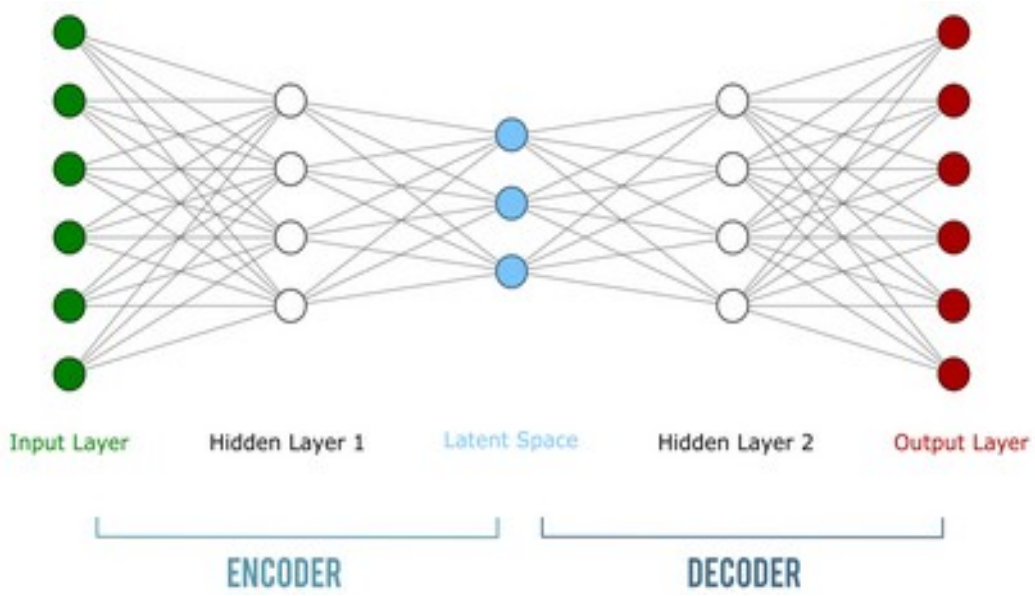
CNN [2]	…			

Anomaly brach

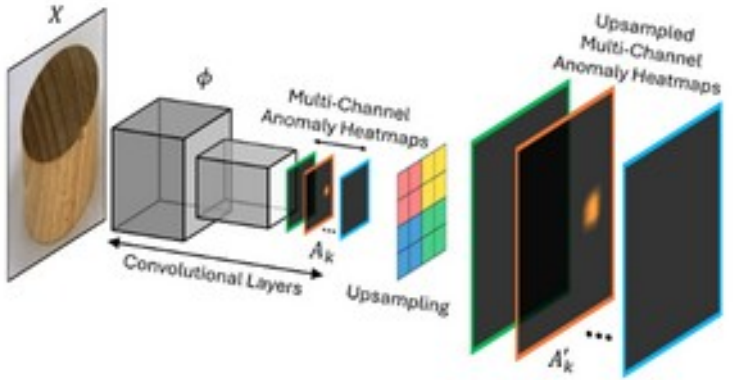
Models

ADStrobot [5]



- Lightweight U-style net
- Spiking network

MultiTypeFCDD [6]



- Standar Convolutional encoder
- Heatmap output to find the Interference

[5] Ritirato, P., Sorrentino, G., & Conficconi, D. (2024). Spiking autoencoder architecture for anomaly detection on satellite data with FPGAs [Poster]. European Space Agency. https://indico.esa.int/event/531/contributions/10627/attachments/6642/11819/poster_SEFUW_print.pdf

[6]George, A., Mihaylova, L., & Anderson, S. (2025). Explainable deep convolutional multi-type anomaly detection. arXiv.https://arxiv.org/abs/2511.11165v1

Anomaly branch

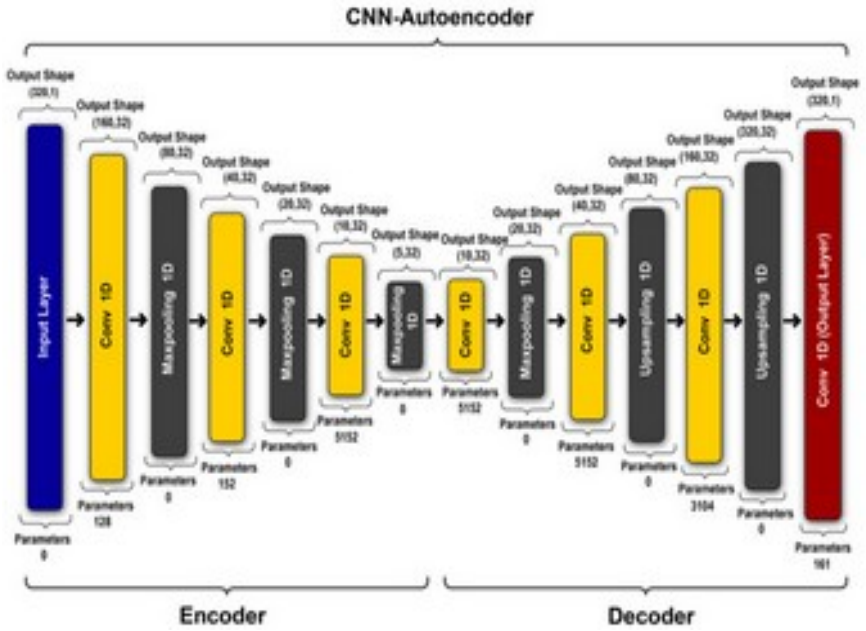
Results

Method	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
DDPM [1]	…			
CNN [2]	…			

Anomaly brach

Proposed approach

Spiking CNN autoencoder [7] (la cit è per l'autoencoder)



- Bello perché 1
- Bello perché 2

Metti qui le references in stile APA, una sotto l'altra se più di una, davanti ad ognuna metti il numeretto [x]

Anomaly branch

Results

Method	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
DDPM [1]	…			
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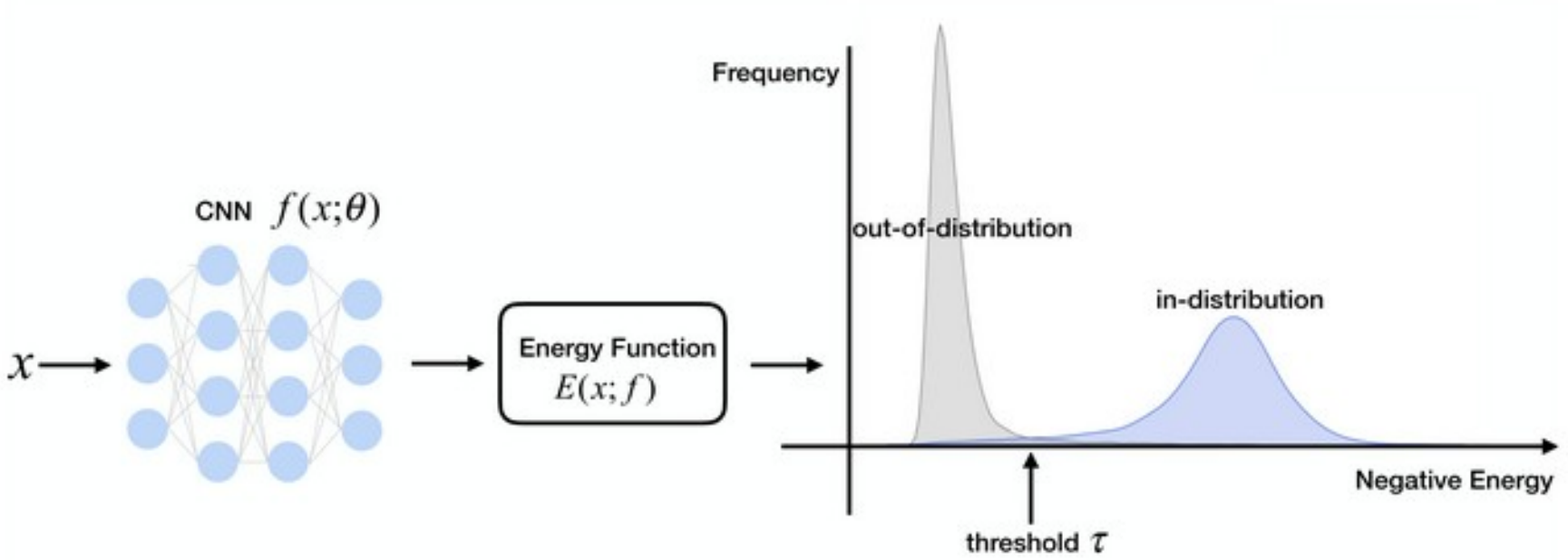
Next steps

Energy

- Very useful in OOD detection [7]
- Can use previously trained models
- Checks the difference in logits betwen in and out of distributions samples



[7]Liu, W., Wang, X., Owens, J. D., & Li, Y. (2021, April 26). Energy-based out-of-distribution detection. ArXiv https://arxiv.org/pdf/2010.03759